Data Base System

ASSIGNMENT: 2

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Assignment No:2

Database Project Proposal:

Hotel Reservation Management System **(HRMS)**

Project Description

The **Hotel Reservation Management System** is designed to computerize and streamline hotel operations such as room booking, customer management, payment, service and feedback

The system replaces manual record-keeping with a centralized database that ensures **accuracy, efficiency, and secure data management**.

It helps hotel staff manage reservations efficiently, avoids double bookings, tracks payments, and improves customer satisfaction by maintaining complet customer and service records. Management can also generat reports for better decision-making.

Core benefits:

- Faster reservation processing
- Reduced human errors
- Centralized and structured records
- Improved customer experience

- Better reporting and monitoring

Target Users:

The main users of the system are:

- Hotel Administrator
- Receptionist
- Hotel Manager
- Customers (indirectly, through staff)
- Hotel Employees (housekeeping, service staff)

Functionalities:

1. Add new hotel details
2. Add, update, or delete rooms
3. Define room types (single, double, deluxe, etc.)
4. Register new customers
5. Update customer information
6. Search customers by ID or name
7. Make room reservation
8. Modify existing reservation
9. Cancel reservations
10. Check room availability
11. Assign rooms to reservations

12. Record check-in detail
13. Record check-out details
14. Generate bills for reservations
15. Process customer payments
16. Store payment methods
17. Manage hotel employees
18. Assign employees to reservations/services
19. Add hotel services (laundry, food, cleaning)
20. Request room services
21. Track service usage
22. Record customer feedback
23. View reservation history
24. Generate daily/monthly reports
25. View payment reports
26. Prevent double booking of rooms
27. Maintain data security and integrity

Database Objects (Tables):

1. **Hotel**
   (hotel_id, name, location, contact_number)
2. **Room**
   (room_id, room_number, status, hotel_id, room_type_id)
3. **Room_Type**
   (room_type_id, type_name, price_per_night, capacity)

4. **Customer**
   (customer_id, name, contact_number, email, address)
5. **Reservation**
   (reservation_id, check_in_date, check_out_date, customer_id, room_id, employee_id)
6. **Payment**
   (payment_id, amount, payment_date, payment_method, reservation_id)
7. **Employee**
   (employee_id, name, role, contact_number, hotel_id)
8. **Service**
   (service_id, service_name, service_cost)
9. **Room_Service**
   (room_service_id, reservation_id, service_id, quantity)
10. **Feedback**
    (feedback_id, rating, comments, customer_id, reservation_id)

Data Relationships:

- One **Hotel** can have many **Rooms** (1-to-Many)
- One **Room Type** can be assigned to many **Rooms** (1-to-Many)
- One **Customer** can make many **Reservations** (1-to-Many)

- One **Room** can be linked to many **Reservations** over time (1-to-Many)
- One **Reservation** has exactly one **Payment** (1-to-1)
- One **Employee** can handle many **Reservations** (1-to-Many)
- **Reservations** and **Services** have a Many-to-Many relationship resolved using **Room_Service**
- One **Customer** can give multiple **Feedback** entries (1-to-Many)